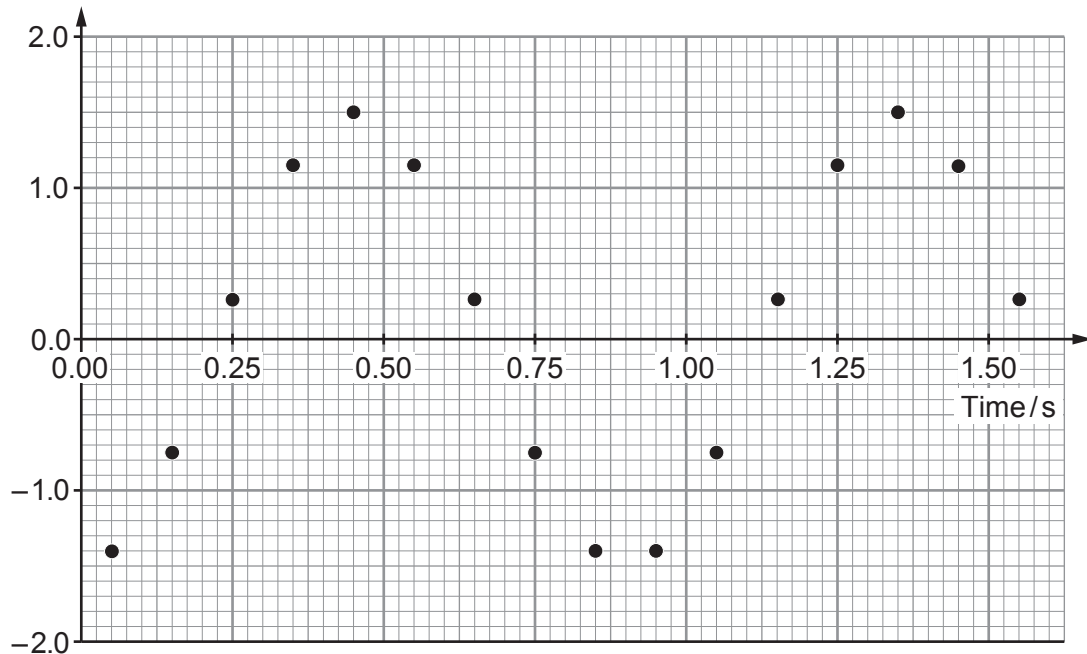


4. A 0.030 kg mass is suspended from a light, vertical spring. A student sets the mass to oscillate vertically. The position of the oscillating mass is tracked using a datalogger. Measurements, taken over a time interval of approximately 1.50 s, are shown on the graph below. The amplitude during this interval was 1.5 cm.

Displacement/cm



- (a) **Draw on the grid** above the curve of best fit. [2]

- (b) Explain what is meant by the 'period of the oscillation'. Use the graph to support your answer. [2]

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- (c) **Complete** the expression for the displacement, y , of the mass. [2]

$$y = 1.5 \cos \left(\left(\frac{2\pi}{\dots\dots\dots} \right) t \dots\dots\dots \right) \text{ cm}$$



- (d) (i) By drawing a tangent to the displacement-time curve, show that the maximum velocity of the oscillating mass is approximately 10 cm s^{-1} . [3]

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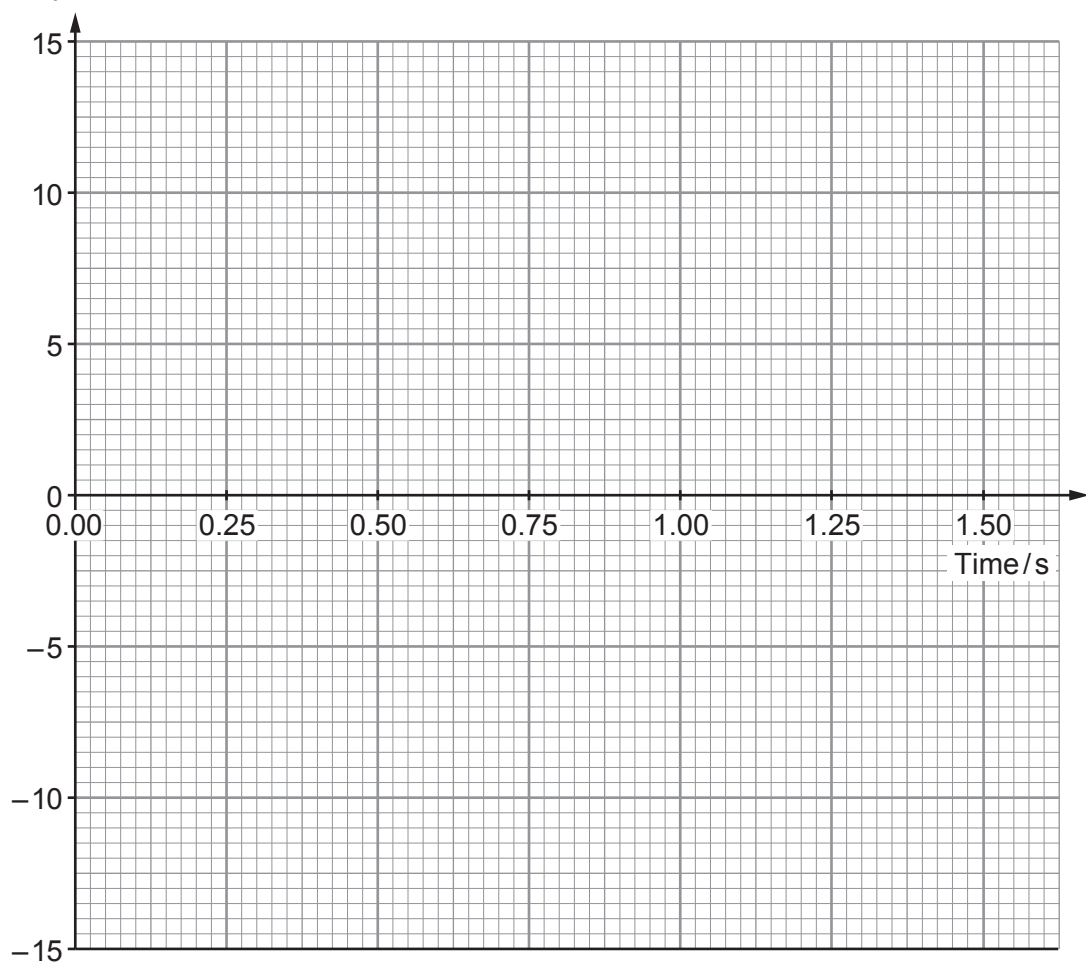
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- (ii) **Sketch** the velocity-time graph for the oscillating mass on the grid below. [3]

Velocity/ cm s^{-1}



(e) Catrin suggests that the 0.030 kg mass is oscillating with simple harmonic motion.

(i) State what is meant by 'simple harmonic motion'. [2]

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(ii) Calculate the value of the spring constant. [3]

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- (f) Catrin also claims that the maximum elastic potential energy stored by the spring is far greater than the maximum kinetic energy of the mass. Evaluate whether or not this claim is correct.

Amplitude of the motion = 1.5 cm

Mass = 0.030 kg

Maximum speed = approximately 10 cm s^{-1}

[6]

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